In demand function,

$Q_d = 100 - 5\,P = 100 - 5(10) = 50$

Hence at

$P = 10, Q_d = 50, Qs = 50.$

Quantity demanded is equal to supply at 50 units when price is ₹10

Example: 12.4

The market demand curve is given by D = 50 – 5P. Find the maximum price beyond which nobody will buy the commodity.

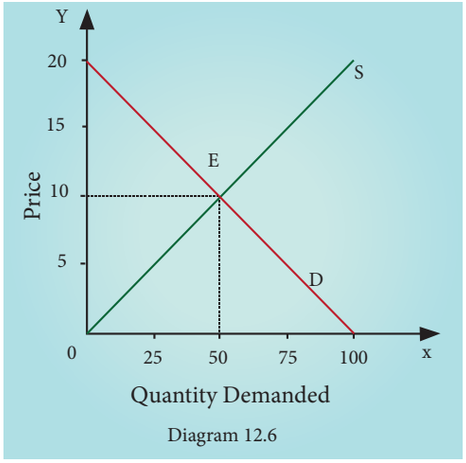


Diagram 12.6

Solution:

Given

$Q_d = 50 - 5P$

$5P = 50 - Q_d$

$5P = 50$ when Q_d is zero.

$$P = \frac{50}{5}$$

P = 10 When P = 10, Demand is 0

Hence P = 10, which is the maximum price beyond which nobody will demand the commodity.

Example: 12.5

The demand for milk is given by

Price (Y)	1	2	3
Demand (X)	100	50	0

Find the linear demand function and its slope.

Solution:

Equation of demand function joining two data points (100, 1) and (50, 2) are (x_1, y_1) and (x_2, y_2) respectively.

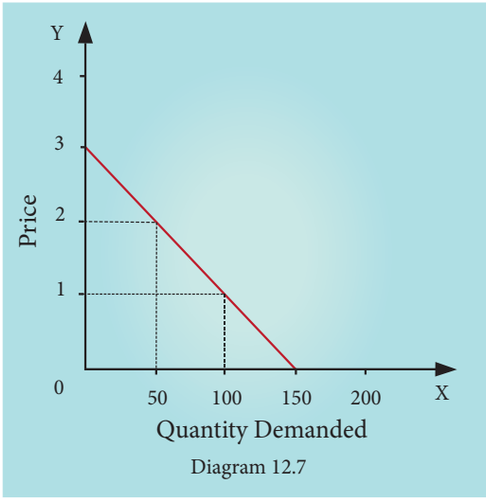


Diagram 12.7

$$\frac{Y - Y_1}{Y_2 - Y_1} = \frac{X - X_1}{X_2 - X_1}$$

$$\frac{Y - 1}{2 - 1} = \frac{X - 100}{50 - 100}$$

$$\frac{Y - 1}{1} = \frac{X - 100}{-50}$$

$$-50\,(Y - 1) = 1\,(X - 100)$$

$$-50Y + 50 = X - 100$$

$$-50Y + 50 + 100 = X$$

$$-50Y + 150 = X$$

$$X = 150 - 50Y$$

Hence the demand function is

$Q_d = 150 - 50P$ and Slope m = – 50



Think and Do for Water Management in your area

Try to find the demand function for water in your street and the daily total demand for water in litre for all purposes.

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